

Caravel M.V.PCB

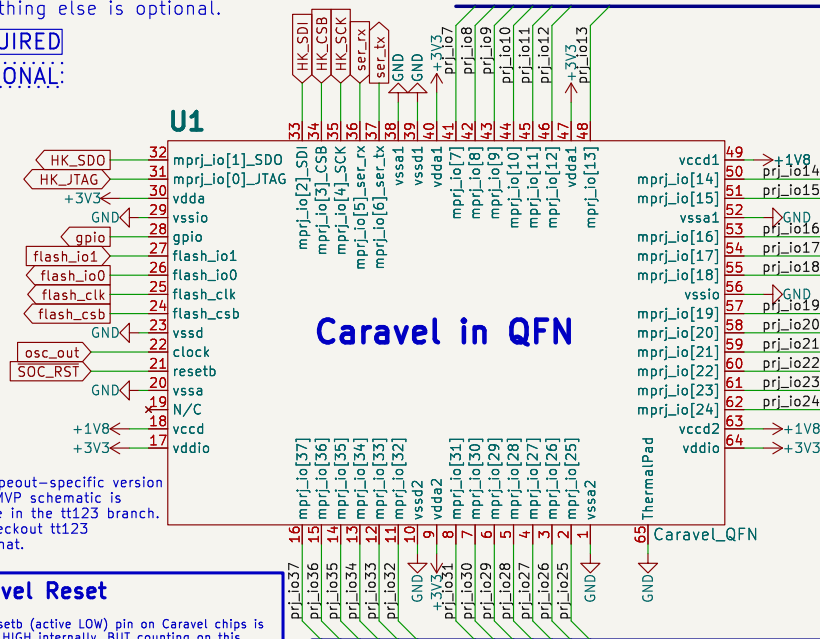
Sample of a minimum viable PCB for ASICs with Caravel on QFN. The REQUIRED support elements are:

- * power: regulated 3v3 and 1v8
- * flash: some memory for executable
- * osc: a CMOS clock signal
- * a bit of reset management

and a way to access the HK SPI is a good idea, or some other way to program the flash (note that there will be SPI line contention if you try to access directly).
Everything else is optional.

REQUIRED

OPTIONAL:



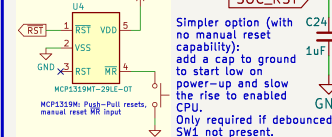
Caravel in QFN

A TinyTapeout-specific version of this MVP schematic is available in the tt123 branch. git checkout tt123 to use that.

Caravel Reset

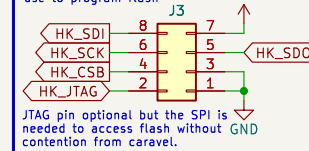
The resetb (active LOW) pin on Caravel chips is pulled HIGH internally, BUT counting on this is glitchy and dangerous. Caravel reference designs use a MCP1319M, as shown here, as a voltage supervisor to manage this and allow debounced reset switch.

Caravel ref design

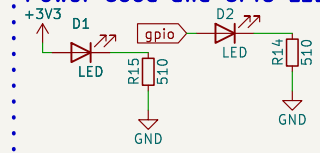


Housekeeping SPI

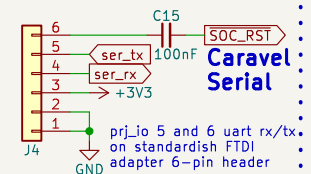
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prj_io 0 to 4, special purpose,
use to program flash
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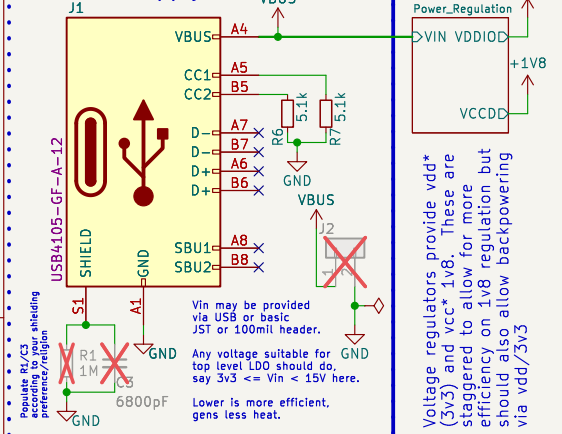
- Power Good and GPIO LED



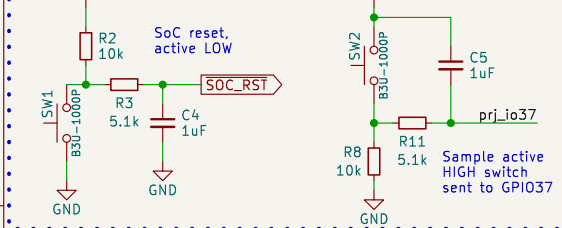
Caravel
Serial



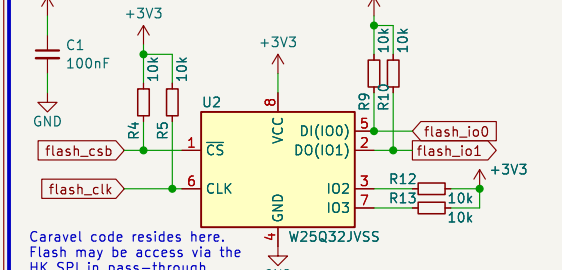
Power Supply



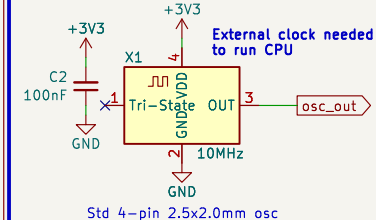
Debounced Switches



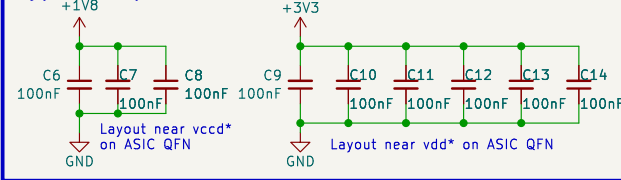
Flash Memory



CMOS Clock Osc



Bypass Caps



- Fids for PnP

- FID1
Fiducial
- FID2
Fiducial
- FID3
Fiducial

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Psychogenic Technologies INC

Sheet: /

File: caravel-mvp.kicad_sch

Title: Caravel Minimum Viable PCB Example

Size: A4	Date: 2023-10-27
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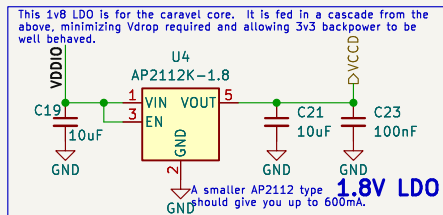
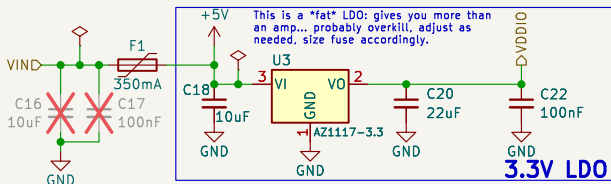
KiCad E.D.A. 8.0.9

Rev: 1.2

Id: 1/2

Voltage Regulators

Simple voltage regulation for logic and core. In a distinct sheet to allow you to easily do fancy stuff, like use switchers or whatever is needed.



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Sheet: /Power_Regulation/

File: power_reg.kicad_sch

Title: Voltage Regulation

Size: User Date: 2023-09-30

KiCad E.D.A. 8.0.9

Rev: 1.1

Id: 2/2